

Certainly! To generate a Makefile rule for creating a Beamer presentation, you can use the Pandoc command switches mentioned earlier. Here's how you might structure your Makefile rule:

```
# Define the input and output directories
INPUT_DIR = ../presentations
OUTPUT_DIR = ./pdfs

# Ensure the output directory exists
$(shell mkdir -p $(OUTPUT_DIR))

# Figure out how to set type to beamer
%.pdf: $(INPUT_DIR)/%.md
    pandoc $< \
        --lua-filter=include-md.lua \
        --citeproc \
        --number-sections \
        -V beamer.theme=Berlin \
        -o $(OUTPUT_DIR)/$@

# Example usage:
# make presentation.pdf
```

Explanation:

1. Directory Paths:

- INPUT_DIR points to the directory containing your Markdown files.
- OUTPUT_DIR points to the directory where you want to save the generated PDFs.

2. Ensure Output Directory Exists:

- The `$(shell mkdir -p $(OUTPUT_DIR))` command creates the output directory if it doesn't already exist.

3. Pandoc Command:

- `$<` refers to the first prerequisite (the Markdown file).
- `--lua-filter=include-md.lua`: This filter is used if you need to include external Markdown files within your main presentation.
- `--citeproc`: Enables citation processing with BibTeX or other reference management tools.
- `--number-sections`: Numbers all section headings.
- `-V beamer.theme=Berlin`: Sets the Beamer theme. You can change this to any valid Beamer theme name as needed.
- `-o $(OUTPUT_DIR)/$@`: Specifies the output file path and name in the OUTPUT_DIR.

Example Usage:

To generate a PDF from your Markdown file, simply run:

```
make presentation.pdf
```

This will create a file named `presentation.pdf` in the `pdfs` directory. Adjust the `beamer.theme` variable as needed for different themes.

Feel free to modify this rule according to your specific requirements!